

to A0 pin of the Arduino board. VCC and GND pins of the sensor are connected to the Arduino's 5V and GND pins, respectively.

Software

Three sets of code are prepared for the system. The first is for the Arduino Uno, which collects sensor data and transmits it via the serial port. The second and third are for HTML and JavaScript, respectively. The HTML code defines the user interface, including the graph and control buttons, while the JavaScript code manages the serial port connection, processes the incoming data, displays it on the page and draws a graph, and allows users to download logged data in CSV format.

To begin, upload the Arduino code by selecting the appropriate board and port in the Arduino IDE. Fig. 5 provides a snippet of the Arduino code.

Web application setup

The web application comprises an HTML page with embedded JavaScript to:

- Read data from the Arduino using the browser's serial API

- Display sensor data in real time

- Log data to a CSV file

The HTML code, shown in Fig. 6, defines the application interface. Save this as index.html, or download it using the QR code provided.

Adding interactivity with JavaScript

The JavaScript code enables communication with the Arduino and provides functionality for real-time graphing and CSV data logging. The key tasks performed include:

- Connecting to the Arduino.** Using the serial API to request and open a serial port

- Reading sensor data.** Parsing the incoming data stream and updating display values

Graphing real time data.

Using Chart.js to dynamically plot moisture levels

- Logging data to CSV.** Storing incoming values in an array and generating a downloadable CSV file

Save the JavaScript code shown

in Fig. 7 as script.js in the exact directory as the HTML file.

Testing the application

To test the complete system:

- Connect the soil moisture sensor to the Arduino and connect the

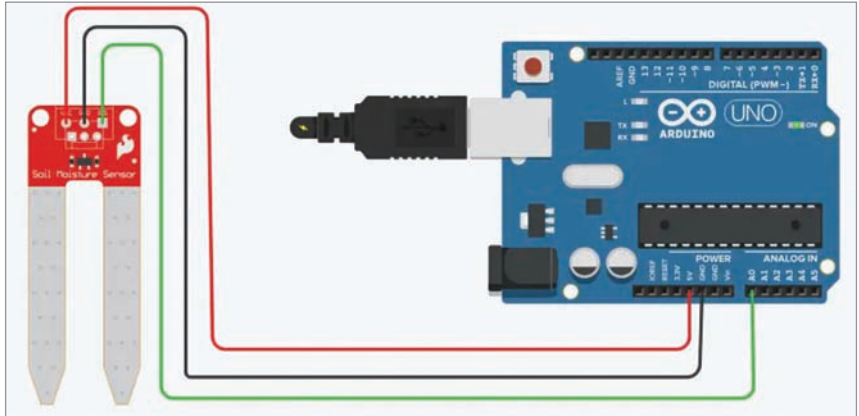


Fig. 3: Physical connection

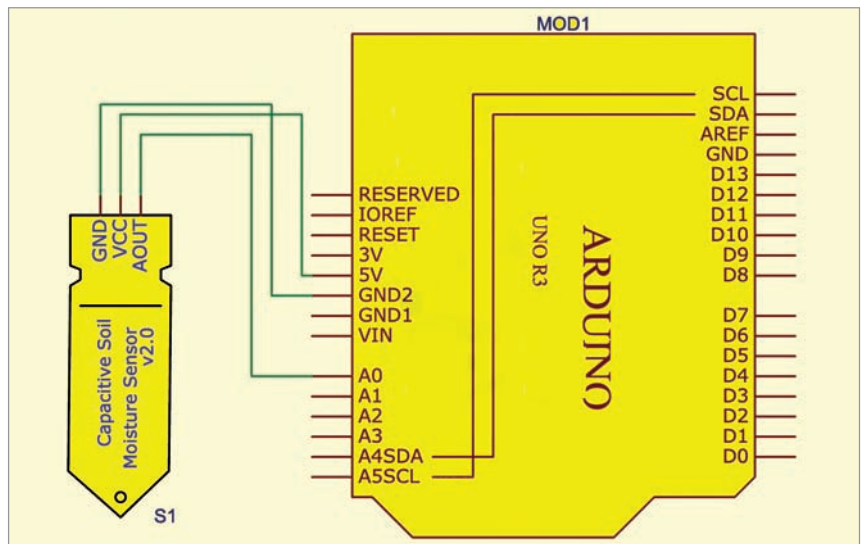


Fig. 4: Circuit diagram

```

sketch_mar18a.ino
1  void setup() {
2      Serial.begin(9600); // Start the serial communication
3  }
4
5  void loop() {
6      int moistureValue = analogRead(A0); // Read moisture sensor value
7      Serial.println(moistureValue);      // Send value to the serial port
8      delay(1000);                        // Delay for 1 second
9  }
10
11

```

Fig. 5: Snippet of Arduino code